Regional Favoritism in Later Periods

Nightlights and Pollution in Leader Birth Regions

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Motivation

- Hodler & Raschky (2014): political leaders direct economic activity toward their **birth regions**, visible in subnational nightlights.
- Recent literature increasingly uses **post-2005 annual windows** and the **harmonized DMSP–VIIRS** panel.
- These windows overlap with global pollution measures (NO_2, $PM_{2.5}$).

$\Rightarrow$ Does the canonical birth-region reduced form still appear in later windows?

What This Paper Does

1. Documents **post-2005 attenuation** of the birth-region nightlights effect:
   - across a window grid, and
   - across theory-favored subsamples (geography, tenure, autocracy, headroom).
2. Tests a **green favoritism** hypothesis — did favoritism shift from *quantity* to *composition* (cleaner growth)?
3. Situates findings against recent replications (Bora 2025; Düben, Hodler & Raschky 2026): pooled long samples may mask later-period attenuation.

Data

Global ADM2 district-year panel combining:

- **PLAD**: leader identity, spell timing, subnational birthplace.
- **Nightlights**: DMSP 1992–2013; harmonized DMSP–VIIRS 2005–2019.
- **NO_2**: ACAG surface NO_2, 2005–2019.
- **$PM_{2.5}$**: ACAG SatPM, longer-window extension.
- **Geography**: GADM ADM2 boundaries.

Treatment ($\text{BirthRegionLeader}_{ict}$): district i in country c is the birth region of the leader in office in year t. Overlapping spells truncated.

Empirical Specification

$$\log(Y_{ict} + 0.01) \;=\; \beta\,\text{BirthRegionLeader}_{ict} \;+\; \alpha_i \;+\; \gamma_{ct} \;+\; \varepsilon_{ict}$$

- α_i: ADM2 fixed effects $\quad \gamma_{ct}$: country-year fixed effects
- β identified from variation *within* country-year, across districts.
- Standard errors clustered by **leader spell** (baseline); country-clustered as robustness.
- Report lag-0, 1, 2; main text focuses on lag-0.

Pollution-intensity outcome:

$$\log(\text{Pollution} + 0.01) - \log(\text{Nightlights} + 0.01).$$

Negative β would indicate *cleaner* birth-region growth.

The Long DMSP Replication Holds

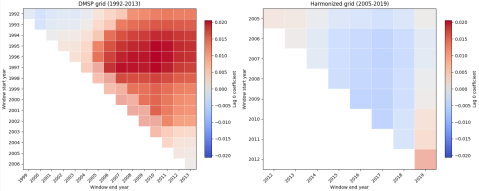
Window	Coef.	SE	p-value
1992–2013	0.0128	0.0059	0.030
1995–2013	0.0187	0.0055	0.001
2000–2013	0.0110	0.0058	0.056
black!7 2005–2013	0.0017	0.0058	0.768
black!7 2005–2019	0.0004	0.0087	0.966

- Long DMSP: $\approx 1.3\%$ activity premium in birth regions, in line with original literature.
- Later windows (shaded): near zero and statistically uninformative.

Attenuation Is Broad: Window-Grid Evidence

- Each cell: a separate ADM2 + country-year FE regression on a (start, end) window.
- DMSP grid: a **block of positive estimates** concentrated in earlier starts and late-2000s/early-2010s ends.
- Harmonized 2005–2019 grid: **small and uninformative throughout.**

⇒ Not a single bad endpoint — a *period-wide* attenuation.



Diagnostics: Why Does the Effect Attenuate?

We narrow, but do not pin down, the explanation:

- **Not a harmonization artifact.** VIIRS-only 2018–2023: lag-0 $= 0.0038$ ($p = 0.722$).
- **Political composition** shifts only modestly.
- **Shorter spell support**: avg. active years per spell 5.22 (1992–2013) vs. 3.97 (2005–2013) — partly mechanical.
- **Headroom**: median treated DMSP DN $= 16.84$ vs. untreated $= 8.83$; treated upper tail near the saturation ceiling.
- **Geocoding quality** stable across periods.

No Post-2005 Subsample Restores the Effect

Long DMSP (1992–2013):

- Africa: 0.0428 ($p = 0.009$); Sub-Saharan Africa: 0.0442 ($p = 0.011$).
- Non-SSA: 0.0053 ($p = 0.370$). *Effect lives where the original literature said it would.*

Post-2005:

- Africa, SSA in DMSP 2005–2013: *negative* and insignificant.
- Long-tenure autocrats; drop top 10% luminosity; high baseline NO_2 countries: all near zero.
- Most suggestive: Harmonized 2005–2019 SSA $= 0.0315$ ($p = 0.135$) — still below our minimum-support threshold ($\beta \geq 0.005$, $p < 0.10$).

Did Favoritism Shift to Cleaner Growth?

Green favoritism: pollution intensity should *fall* in birth regions even when nightlights do not rise.

Outcome (2005–2019, lag 0)	Coef.	SE	p
$\log(\text{Nightlights} + 0.01)$	-0.0026	0.0091	0.778
$\log(NO_2 + 0.01)$	0.0308	0.0154	0.045
NO_2 pollution intensity	0.0333	0.0179	0.063
$PM_{2.5}$ intensity (lag 0)	0.0041	0.0095	0.665

- Pollution coefficients are zero to positive — not negative.
- Africa/SSA intensity points are negative but imprecise (cannot carry the story).
- NO_2 loses precision under country clustering: read as *suggestive against* green favoritism.

Takeaways

- Post-2005 attenuation of the birth-region nightlights effect is **broad**: window grid, geography, tenure, autocracy, headroom, pollution support.
- Long DMSP replication **still holds**, especially in Africa / Sub-Saharan Africa.
- Pollution evidence **rules out** a simple shift toward greener birth-region growth.
- Historical 95% CIs contain the post-2005 points $\Rightarrow$ *loss of detectable signal*, not proven absence.
- Diagnostic contribution: long pooled samples may mask period-specific attenuation.

Thank You

Questions?

We thank Professor Joshua Blumenstock for feedback.